

Draft - expert review pending

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Unit 4 assessment

Unit summary

Each workplace sector, industry, construction, agriculture and services, carries a characteristic hazard portrait, and incidents grade from near-miss through injury and fatality to the under-counted chronic illness; the 2012 Karachi factory fire shows small failures lining up behind a locked door. An OSH management system runs the PDCA loop, plan, do, check, act, with leading indicators preferred over injury counts, under the international reference points of ILO Convention 155 and ISO 45001. Regulation supplies the ladder of act, regulations, codes and guidance; internal control supplies the layers, philosophy, policy, procedure, supervision, records, that make rules real, and the philosophy layer decides the rest. The duties triangle, employer, worker, government, carries the rights, know, protect, refuse, raise, and the OHS professional works the six- role web while the employer holds the duty. The incident pyramid and the Swiss cheese model explain why small failures and serious harm share causes. Pakistan's statutes layer the 1997 Environmental Protection Act with its NEQS, the Factories Act 1934 and provincial law, with much competence devolved since 2010; an HSE plan turns them into one page with names and dates. The quiet exposures, heat, dust, noise, stress, ergonomics, are managed by ordinary tools mapped to the system's elements.

Summative assessment

Multiple-choice questions (MCQs)

1. In the Faisalabad spinning-mill study, the workers' burden was chiefly:
(Understand) a) falls from height b) respiratory disease with lost work days
c) hearing loss d) pesticide poisoning

2. The 2012 Karachi factory fire is described as "ordinary in its mechanism" because it: (*Understand*) a) was caused by an unusual chemical reaction b) resulted from several small failures lining up c) happened in an old building only d) was deliberately set
3. The four stages of an OSH management system's engine are: (*Remember*) a) predict, decide, correct, approve b) plan, do, check, act c) policy, drill, count, audit d) scope, hazard, control, record
4. On the regulatory ladder, codes of practice sit between: (*Understand*) a) acts and regulations b) regulations and guidance c) guidance and company rules d) statutes and the constitution
5. Under the duties triangle, the employer's core duty is: (*Remember*) a) following the safety officer's advice in all cases b) providing a workplace without avoidable harm c) reporting every near-miss to government d) paying for the worker's own protective equipment
6. In the incident pyramid, the widest band at the base is: (*Remember*) a) fatalities b) serious injuries c) near-misses and unsafe acts d) property damage
7. A director claims two accident-free years prove safety. The strongest reply uses: (*Apply*) a) the factory's equipment age b) the distinction between leading and lagging indicators c) the number of safety officers employed d) the size of the insurance policy
8. After the 18th Constitutional Amendment, a workplace's legal homework grew because: (*Understand*) a) national law was abolished b) labour and environmental competence largely devolved to the provinces c) ILO conventions became binding directly d) the courts stopped hearing safety cases
9. Which pairing of tool and system element is correct? (*Apply*) a) near-miss book and plan b) sound meter and act c) legal register and plan d) corrective-action tracker and do
10. The WHO's estimate of 12 billion working days lost yearly belongs to the challenge of: (*Understand*) a) heat b) dust c) ergonomics d) stress and mental health

Restricted-response questions (RRQs)

1. Name the four workplace sectors and one characteristic hazard of each.
(*Remember*)
2. Explain why chronic illness is the incident grade that statistics under-count.
(*Analyze*)
3. Why is a system steered only by injury counts "steered by its failures"?
Name the alternative indicator type. (*Analyze*)
4. List the five layers of internal control, and name the one the others inherit from. (*Understand*)
5. Name the four worker rights and explain why the right to refuse needs the others to work. (*Understand*)
6. What does the shape of the incident pyramid tell a safety officer to count, and why? (*Understand*)
7. Explain the Karachi fire in Swiss cheese terms, in two sentences. (*Analyze*)
8. State the six steps of establishing an HSE plan, in order. (*Remember*)
9. Give two mechanisms by which the external environment crosses into a workplace. (*Apply*)
10. Match four HSE tools to the management-system element each supports.
(*Apply*)

Extended-response questions (ERQs)

1. Your cousin is joining a spinning mill. Using the sector portrait and the PDCA cycle, write what the mill should already have in place for dust, from policy to review, and the one question your cousin should ask on the first day.
(*Analyze*)
2. A safety officer's report is ignored by the owner. Using internal control's layers and the duties triangle, analyse what has failed, who still holds the duty, and what the officer's remaining roles are. (*Evaluate*)
3. A director says, "No accidents for two years, so we are safe." Write the reply using the incident pyramid, the leading-versus-lagging distinction and the organizational environment, ending with two measurements to start this month. (*Evaluate*)
4. Write the one-page HSE plan outline for your practicum school's sports day: scope, five hazards across three classes, one measurable objective per

hazard with its hierarchy level, the training line, and the monitoring line with names and times. *(Create)*

5. Integrative: the school's June examination series is the workplace. Write the full HSE plan outline for it, covering the quiet health exposures, heat, noise and stress, the external environment that feeds them, and the toolbox, naming actual tools, that makes each measurable, with the monitoring and review lines to close the loop. *(Create)*

Answers and marking guidance

MCQ answer key

1. b - the study found respiratory disease and work absence in Faisalabad's spinning mills.
2. b - locked exits, no alarm, stored combustibles and no audit lined up; each alone is ordinary.
3. b - plan, do, check, act, the loop borrowed from quality management.
4. b - regulations say how; codes say what counts as doing it well; guidance advises.
5. b - the duty of care: a workplace without avoidable harm, with information, training and equipment.
6. c - near-misses and unsafe acts are the base; the serious injury is the apex.
7. b - injury counts are lagging; two quiet years may mean no reporting, not no failures.
8. b - devolution moved labour and much environmental competence to the provinces.
9. c - the legal register feeds planning; the near-miss book and meters feed checking; the tracker feeds acting.
10. d - depression and anxiety cost an estimated 12 billion working days a year.

RRQ model answers and mark schemes

1. Model: industry, dust and unguarded machines; construction, falls and excavation collapse; agriculture, pesticides and machinery; services, lifting, stress and infection. **Marks (4):** 1 per sector-hazard pair.

2. Model: fatality and injury statistics count events; chronic illness develops slowly, often after leaving the job, and is attributed to the person rather than the exposure, so it misses the registers, as the uncle's cough shows. **Marks (2):** 1 for the slow-development point; 1 for the attribution point.
3. Model: injury counts record what already went wrong, so the system reacts after harm; leading indicators, inspections done, near-misses reported, measure what the system is doing before harm. **Marks (2):** 1 for the reactive point; 1 for naming leading indicators.
4. Model: philosophy, policy, procedure, supervision, records; the others inherit from philosophy, what leadership actually rewards. **Marks (3):** 2 for all five layers; 1 for naming philosophy as the parent.
5. Model: know, protect, refuse, raise; refusal without the right to know and to raise concerns is blind and punished, so the rights work as a set. **Marks (3):** 2 for four rights; 1 for the interdependence argument.
6. Model: count the base, near-misses and unsafe acts, because they share causes with the apex, so the base is where the next fatality is visible and cheapest to fix. **Marks (2):** 1 for what to count; 1 for the shared-causes reason.
7. Model: the locked door, missing alarm, stored combustibles and missing audit were four slices with holes; the harm passed when the holes lined up, and any working slice would have stopped it. **Marks (2):** 1 for slices and holes; 1 for the alignment.
8. Model: scope, hazards, objectives, controls and resources, training and communication, monitoring and review. **Marks (3):** all six in order for 3; four or five for 2; two or three for 1.
9. Model: any two of heat crossing into uncooled rooms, road noise and air entering from the street, floods interrupting travel and supply, power cuts stopping fans and pumps. **Marks (2):** 1 per mechanism.
10. Model: legal register to plan; training kit and checklists to do; inspection forms, near-miss book, sound meter to check; review agenda and corrective-action tracker to act. **Marks (4):** 1 per correct pairing.

ERQ rubrics

1. **Analytic rubric.** *Portrait used: dust as the mill's characteristic exposure (0-2, Understand) · PDCA stages each given a dust content (0-4, Analyze) · The first-day question is real and specific (0-2, Apply) · Clarity (0-2).* Total 10.

2. **Analytic rubric.** *Layers analysed: which failed and which remain (0-3, Analyze) · Duty correctly located with the employer (0-2, Understand) · Officer's remaining roles from the six-role web (0-3, Apply) · Clarity (0-2). Total 10.*
3. **Analytic rubric.** *Pyramid applied to the quiet years (0-3, Analyze) · Leading-versus-lagging correct (0-2, Understand) · Two concrete measurements with dates (0-3, Apply) · Clarity (0-2). Total 10.*
4. **Analytic rubric.** *Six steps present (0-2, Apply) · Hazards span three classes, correctly named (0-3, Understand) · Objectives measurable with hierarchy levels (0-3, Analyze) · Monitoring line with names and times (0-2). Total 10.*
5. **Analytic rubric.** *Quiet exposures identified and named as challenges (0-3, Understand) · External-environment links shown (0-2, Analyze) · Tools concrete and matched to system elements (0-3, Apply) · Monitoring and review close the loop (0-2, Create). Total 10.*

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